

can enforce a higher degree of security. If that bank has one million customers then the records of their transactions will be stored on those one million devices. In fact, a hacker will have to hack one million devices in real-time rather than a single server. This is the major advantage of using the decentralised blockchain technology.

In case of a centralised application, if a hacker penetrates the server of a bank, then the details and records of all the customers can be copied. That is the main reason why government agencies should focus on decentralising their Web based applications.

Using Blockchain technology, government servers used for land registry, citizen information (including Aadhaar), Permanent Account Number (PAN) and many others can be secured using decentralised apps.

The blockchain based decentralised applications can be used for the following:

- Birth, marriage and death certificates
- Asset and land registry
- Digital identity of government documents
- Incorporation services
- Notarised documents
- Personalised government services
- Social welfare and benefits
- Taxation
- Polling, voting at assembly elections, etc

Decentralised application (dApp) with the secured blockchain

The decentralised application (dApp) executes on the distributed channels so that hacking the app is next to impossible. A traditional centralised application is deployed on a single server. The main limitation with this approach is that if the central server is hacked then everything can be damaged or copied from it. In case of decentralised applications, there is no single server but rather storage is done on all the client devices so that the replication of the transaction can be done with maximum availability of transaction records. In the case of a decentralised application, the hacker will have to crack all the devices associated with that application and that will be very difficult in real-time using smart contracts. In a smart contracts based dApp, dynamic token sharing is implemented so that the transactions will have the maximum security measures.

Smart contracts

Smart contract programming is required for global transactions. This means that transactions can be executed between people who can't communicate because they are on different continents, speak different languages and practice different traditions. Smart contracts automatically validate transactions and business dealings between the people who can't understand each other's languages.

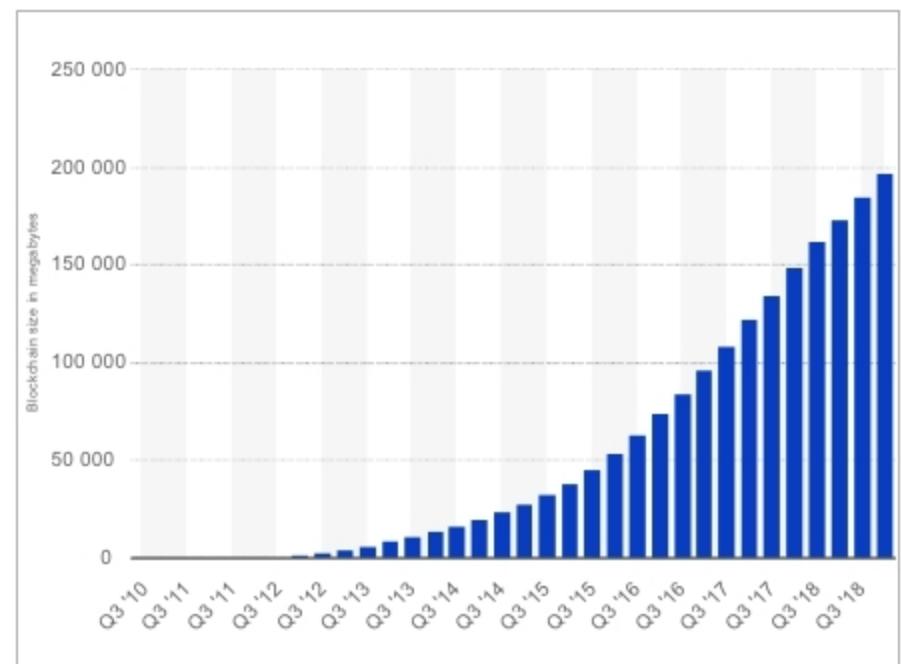


Figure 1: Blockchain size from 2010 to 2018

Free and open source tools for blockchain development

Listed below are some of the leading FOSS tools for blockchain development.

Hydrachain

<https://github.com/HydraChain/hydrachain>

- Users can create permissioned distributed ledgers
- They can set up a private chain
- Fully compatible with the Ethereum protocol

Multichain

<https://www.multichain.com/>

- Compatible with Bitcoin
- Fully customisable
- Fine-grained managed permissions
- Enables the rapid creation and deployment of a new blockchain
- Powerful data sharing and encryption

OpenChain

<https://www.openchain.org/>

- Digitally signed transactions
- Custom rule definitions for ledgers
- Robustness and fine validation
- Client server architecture
- Module design with real-time validations
- Immutability with security

Ethereum

<https://www.ethereum.org/>

- Smart wallets and smart money
- Supports the creation and development of users' own cryptocurrency
- Security against third party intervention or downtime
- Execution of smart contracts
- Virtual shares with crowd fund and crowd sale